



جامعة الشارقة
UNIVERSITY OF SHARJAH
College of Sciences
Department of Mathematics

Course Syllabus

Course Title **Introduction to Probability and Statistics**

Course Code
1440281

Fall Semester, 2023/ 2024

COURSE SYLLABUS

Course Title	Introduction to Probability and Statistics									
Course Code	1440281									
Credit Hours	3									
Course Category		Univ. Req.		Col. Req.	√	Prog. Req.	Others			
Course Type	√		Compulsory					Elective		
Pre-requisite(s)	Calculus 1: 1440131									
Co-requisite(s)	None									
Semester	Spring				Year		2023/ 2024			
Instructors Name	Marwan Al-Momani									
Office Location	M7-112-A									
Tel. No.	1733									
Email	malmomani@sharjah.ac.ae									
Lecture Times										
Office Hours	MW: 12:30-1:30 or by appointment									

Course Description (as in the catalogue):

Descriptive statistics; Axiomatic probability; Random variables and their moments; Special discrete and continuous distributions; Sampling distributions; Estimation; Hypothesis testing; Linear regression; Analysis of variance.

Course Objectives (optional):

This course aims to introduce the students to concepts and methods of descriptive statistics, probability, random variables, sampling distribution, and statistical inference.

Course Learning Outcomes:**On completion of this course, the student will be able to:**

1. Be able to apply statistical techniques to summarize data graphically and numerically.
2. Use probability methods to formalize and solve real-life problems.
3. Solve problems related to discrete and continuous probability distributions.
4. Use statistical techniques to construct Confidence Intervals for the population parameters: mean and proportion.
5. Test hypotheses about some of the population parameters: mean and proportion.
6. Analyze data using R and writing a short report.

Alignment of Course Student Learning Outcomes to Program Student Learning Outcomes

Program SLOs	Course SLOs
Apply a broad range of core mathematical knowledge and techniques, including advanced calculus, linear algebra, geometry, differential equations, probability, and statistics.	1,2,3,4,5,6
Continuously apply scientific experiments and research methods to build on existing knowledge with modern and innovative ideas.	1,2,3,4,5
Utilize mathematical knowledge and IT skills to design, implement and enhance computer programs.	1,6
Function on multidisciplinary teams.	5,6
Analyze statistical data to identify and solve applied science problems.	1,2,4,5,6
Understand professional and ethical responsibility.	1,2,3,4,5,6
Communicate mathematical knowledge directly and indirectly with precision, clarity, and organization.	1,2,3,4,5,6
Formulate mathematical models to solve real-life problems in a contemporary global and societal context.	1,2,3,5

Program SLOs	Course SLOs
Utilize modern techniques and skills to achieve pre-determined goals and improve overall performance in a professional setting.	5,6

Teaching and Learning Resources:

Textbook	Probability & Statistics for Engineering & the Sciences, 8th Ed. 2008, by J. Devore
Supplementary References	Probability and Statistics for Engineering and Scientist, 8 th edition, by Walpole, Myers, Myers, and Ye, 2008. The instructor has developed his own Tutorial questions along with a course summary

Weekly Distribution of Course Topics/Contents

Week	Section(s)	Subject(s)	Suggested Practice Problems	CLO
1st, and 2nd	Chap.1:1-2	Introduction to Stats. Data Representation. Describing center, spread. Association, contingency, intro to correlation and regression. Cautions in analyzing associations. Gathering data.	1-3	1
3rd	Chap. 1: 3-4	Introduction to Stats. Data Representation. Describing center, spread. Association, contingency, intro to correlation and regression. Cautions in analyzing associations. Gathering data.	10-20, 35, 36, 39, 49, 51, 54, 55	1,6
4rd and 5th	Chap.2: 1-2	Probability. Qualification of randomness by probability. Basic definitions of events, trials, and sample space. Independence of trials. Complements. Use of tree and Venn diagrams for visualization. Disjoint events. Intersection, union. Addition and multiplication rules. Conditional probability, contingency tables, independent events.	1, 3, 4, 6, 11, 12, 17, 19, 20, 25, 26	2
6th	Chap.2:2-5	Probability. Disjoint events. Intersection, union. Addition and multiplication rules. Counting Techniques. Conditional probability, contingency tables, independent events.	30, 31, 32, 33, 34, 36, 37, 45, 47, 48, 49, 59, 60, 72, 74	2,6
7th	Chap.3:1-6	Probability distributions. Random Variable. PMF and CDF. Most common discrete distributions (Bernoulli, Binomial, Poisson).	1, 6, 7, 11, 12, 13, 23, 24, 29, 30, 32, 34, 35, 46, 47, 48, 49, 50, 52, 79, 80, 81, 85, 86	3
8th and 9th	Chap.4:1-6	Probability distributions. Random Variable. PDF and CDF.). Most common continuous distributions (Uniform, Exponential, Normal).	1, 2, 5, 7, 8, 11, 12, 13, 28, 29, 32, 33, 36, 37, 59, 60, 61	3
10th	Chap.5: 4	Sampling distributions. Distributions of sample's mean, and sample's variance. The ratio	37, 38, 41, 46, 47, 48, 50, 53	3

	Chap.6: 1-2	of two variances. Student t-distribution. Chi-squared distribution. F-distribution. Central Limit theorem.		
11th	Chap.7:1-2	Confidence intervals. Interval and point estimation of a population parameter. Confidence level and confidence coefficient.	1, 2, 3, 4, 7	4
12th	Chap.7: 3-4	Confidence intervals. Interval and point estimation of a population parameter. Confidence level and confidence coefficient.	2, 3, 4, 5, 6, 7, 10, 13, 14, 16, 17, 28, 29, 30, 36, 37, 38, 39	4,6
13th	Chap.8:1-3	Hypothesis testing. Null hypothesis. Alternate hypothesis. P-value. Power of a test.	1, 2, 3, 4, 5, 6, 7, 8, 10, 15, 16, 17, 18, 19, 21, 37, 38, 40, 41	5
14th and 15th	Chap.8:4-5	Hypothesis testing. Null hypothesis. Alternate hypothesis. P-value. Power of a test.	47, 48, 49, 50, 51, 52, 53	5,6
16th	Chap.9:1-4	Comparing two groups. Independent group t-test. Paired group t-test.	If time permits 36, 40 43	5

Note: This course uses the statistical package R: <https://www.r-project.org>. R output results may show up in exams and quizzes.

Students are assessed as follows:

Assessment Tool(s)	Date	Weight (%)
Semester activities: quizzes and assignments	TBN	30
Midterm Exam	TBN	30
Final Exam	TBN	40
Total		100

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	Competencies
1. Be able to apply statistical techniques to summarize data graphically and numerically.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, and class discussion.	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	2- Oral communication 3- Written communication 4-Teamwork/ collaboration 12-Problem solving 13-Numeracy
2. Use probability methods to formalize and	Lecture-based instruction, assignments, assigned readings,	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and	1-Critical and analytical thinking 4-Teamwork/ collaboration

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	Competencies
solve real-life problems.	problem-solving integrating active learning, and class discussion.		instructor evaluation by students, peer evaluation	12-Problem solving 13-Numeracy
3. Solve problems related to discrete and continuous probability distributions.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, and class discussion.	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 4-Teamwork/ collaboration 5-Intellectual curiosity and lifelong learning 12-Problem solving 13-Numeracy
4. Use statistical techniques to construct Confidence Intervals.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, and class discussion.	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 4- Teamwork/ collaboration 5-Intellectual curiosity and lifelong learning 12-Problem solving 13-Numeracy
5. Test hypotheses about some of the population parameters: mean and proportion.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, class discussion,	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 3-Written communication. 4-Teamwork/ collaboration 12-Problem solving 13-Numeracy
6. Perform ANOVA to test more than two means.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, class discussion,	Quizzes, Midterm and Final exams.	Tests score, post exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 3-Written communication. 4-Teamwork/ collaboration 12-Problem solving 13-Numeracy
7. Compute and interpret the results of simple and multivariate regression. Be able to perform multiple regression using	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, class discussion,	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 3-Written communication. 4-Teamwork/ collaboration

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators	Competencies
computer software.				5-Intellectual curiosity and lifelong learning 12-Problem solving 13-Numeracy
8. Analyze data using R and writing a short report.	Lecture-based instruction, assignments, assigned readings, problem-solving integrating active learning, class discussion,	Quizzes, Midterm, and Final exams.	Tests score, post-exam analysis, Course and instructor evaluation by students, peer evaluation	1-Critical and analytical thinking 3-Written communication. 4-Teamwork/ collaboration 5-Intellectual curiosity and lifelong learning 12-Problem solving 13-Numeracy

Attendance policy:

A Student should attend all classes, laboratory sessions, practical applications, and exams in the course. When the absence of a student reaches 10%, the instructor will give him/her a first warning; if absence reaches 15%, the student will receive a final warning; should the absences reach 20% of the total credit hours, the student will not be allowed to sit for the final examination and fail the course. **Refer to the Academic Bylaws in the University Catalog (2010-2011), Class Attendance section, pp. 22.**

Plagiarism/Cheating:

The University regards most seriously any acts of dishonesty in assessment, such as plagiarism, collusion, cheating, resubmission of previously marked work in different courses, examination misconduct, and theft of other students' work.

Acts of dishonesty in assessment could result in penalties, including failure in the course and possible exclusion from the University. **Refer to the Academic Bylaws in the University Catalog (2010-2011), Academic Violations section, pp. 30-31.**

Examination Policy & Appeals Against Academic Assessment

Midterm and Final exams will be administered at the scheduled time as predetermined by the College of Science. No test scores will be dropped, and no retakes will be offered for poor performance. The course instructor must approve absences on the test days before the scheduled test. An unexcused absence from a test may result in a grade of "0" for that test. All initial contact for appeals against an academic assessment should be via the course coordinator. Students will be required to make an appointment to discuss the appeal. If required, the appeal will be escalated to the Head of the Department of the course, and other bodies as per the University of Sharjah standard operating procedure. **Refer to the Academic Bylaws in the University Catalog (2010-2011), Final Exams and Incomplete Grades section, pp. 20.**

Notes:

Professional Competency Domains			
Knowledge	✓	Ethics & Professionalism	✓
Communication Skills		Lifelong Learning	✓
Policies & Procedures	✓	Research Methodology	✓